Information Systems Management of Operations, Quality and Processes

Group MALT (Mave, Anne, Louis, Thea)

Exercise 1: Derive the logic Diagram

Given precedence relationships

1. `U → G → H`
2. `Start → E`
3. `E → U`
4. `E → I`
5. `P → K` and `R → K`
6. `E → M → T`
7. `O → R`
8. `H → P` and `T → P`
9. `O → R`
10. `I → S → T`
11. `T → P`
12. `I → W → O`
13. `H → P`

The exercise gives us precedence information and asks us to derive the logic diagram representing the complete logical dependency structure given.

Strategy: Derive the dependency structure from the given information ➔ Derive the logic diagram

Precedence Structure: The resulting network can be summarized as follows:

- Start → E → U → G → H
- E → I → S → T

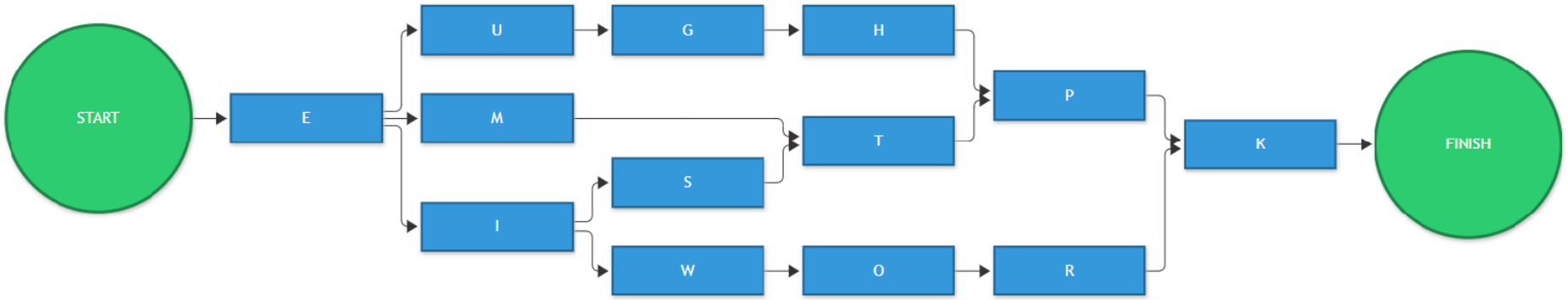
- E → M → T
- I → W → O → R
- H → P
- T → P
- P → K
- R → K
- K → Finish

The resulting logic diagram below identifies the activities that can proceed in parallel and those that must wait for other activities to finish. The network contains three important convergence points:

- **Activity T:** can start only after both M and S have been completed.
- **Activity P:** can start only after both H and T have been completed.
- **Activity K:** can start only after both P and R have been completed.

Repeated relationships from the original exercise have been represented only once.

Logic diagram



Exercise 2 — Critical Path Analysis (CPA) - Milwaukee ES/EF and LS/LF Network

Objective

The exercise asks us to:

1. Add the **Earliest Start (ES)** and **Earliest Finish (EF)** times.
2. Add the **Latest Start (LS)** and **Latest Finish (LF)** times that will not delay the overall project.
3. Identify the **critical path**.

The network contains 12 activities:

A, B, C, D, E, F, G, H, J, K, L, M

1. Forward Pass — ES and EF

The **forward pass** moves from **Start** → **Finish** through the network.

It determines the earliest possible time each activity can start and finish.

Formulas

For the first activity:

$$ES = 0$$

For an activity with one predecessor:

$$ES = EF_{\text{predecessor}}$$

For an activity with multiple predecessors:

$$ES = \max\left(EF_{\text{all predecessors}}\right)$$

Once ES is known:

$$EF = ES + Duration$$

Forward-pass calculations

A

A has no predecessor:

$$ES_A = 0$$

$$EF_A = 0 + 3 = 3$$

B

B follows A:

$$ES_B = EF_A = 3$$

$$EF_B = 3 + 2 = 5$$

C

C starts directly from Start:

$$ES_C = 0$$

$$EF_C = 0 + 10 = 10$$

D

D follows C:

$$ES_D = EF_C = 10$$

$$EF_D = 10 + 5 = 15$$

F

F starts directly from Start:

$$ES_F = 0$$

$$EF_F = 0 + 12 = 12$$

G

G follows F:

$$ES_G = EF_F = 12$$

$$EF_G = 12 + 6 = 18$$

H

H has two predecessors: C and G.

Their EF values are:

$$EF_C = 10$$

$$EF_G = 18$$

Therefore:

$$ES_H = \max(10, 18) = 18$$

$$EF_H = 18 + 11 = 29$$

E

E has three predecessors: B, D and H.

Their EF values are:

$$EF_B = 5$$

$$EF_D = 15$$

$$EF_H = 29$$

Therefore:

$$ES_E = \max(5, 15, 29) =$$

$$EF_E = 29 + 7 = 36$$

K

K starts directly from Start:

$$ES_K = 0$$

$$EF_K = 0 + 5 = 5$$

L

L follows K:

$$ES_L = 5$$

$$EF_L = 5 + 7 = 12$$

M

M follows L:

$$ES_M = 12$$

$$EF_M = 12 + 5 = 17$$

J

J has two predecessors: H and M.

$$EF_H = 29$$

$$EF_M = 17$$

Therefore:

$$ES_J = \max(29, 17) = 29$$

$$EF_J = 29 + 2 = 31$$

The project must wait for both E and J.

Therefore:

$$Project\ Duration = \max(36, 31) =$$

2. Backward Pass — LS and LF

The **backward pass** works in the opposite direction:

Finish → Start

It determines how late each activity can start and finish **without delaying the overall project completion time**.

We begin with the project completion time:

$$Project\ Finish = 36$$

Formulas

For the final activity:

$$LF = Project\ Duration$$

Then:

$$LS = LF - Duration$$

For an activity with one successor:

$$LF = LS_{\text{successor}}$$

For an activity with multiple successors:

$$LF = \min\left(LS_{\text{all successors}}\right)$$

Backward-pass calculations

E

E finishes the main branch:

$$LF_E = 36$$

$$LS_E = 36 - 7 = 29$$

J

J also leads to project completion:

$$LF_J = 36$$

$$LS_J = 36 - 2 = 34$$

H

H is a predecessor of both E and J.

Their LS values are:

$$LS_E = 29$$

$$LS_J = 34$$

Therefore:

$$LF_H = \min(29,34) =$$

$$LS_H = 29 - 11 = 18$$

M

M precedes J:

$$LF_M = LS_J = 34$$

$$LS_M = 34 - 5 = 29$$

L

L precedes M:

$$LF_L = LS_M = 29$$

$$LS_L = 29 - 7 = 22$$

K

K precedes L:

$$LF_K = LS_L = 22$$

$$LS_K = 22 - 5 = 17$$

G

G precedes H:

$$LF_G = LS_H = 18$$

$$LS_G = 18 - 6 = 12$$

F

F precedes G:

$$LF_F = LS_G = 12$$

$$LS_F = 12 - 12 = 0$$

D

D precedes E:

$$LF_D = LS_E = 29$$

$$LS_D = 29 - 5 = 24$$

C

C has two successors: D and H.

Their LS values are:

$$LS_D = 24$$

$$LS_H = 18$$

Therefore:

$$LF_C = \min(24, 18) = 18$$

$$LS_C = 18 - 10 = 8$$

B

B precedes E:

$$LF_B = LS_E = 29$$

$$LS_B = 29 - 2 = 27$$

A

A precedes B:

$$LF_A = LS_B = 27$$

$$LS_A = 27 - 3 = 24$$

3. Summary Table

Activity	Duration	ES	EF	LS	LF	Float
A	3	0	3	24	27	24
B	2	3	5	27	29	24
C	10	0	10	8	18	8
D	5	10	15	24	29	14
E	7	29	36	29	36	**0**
F	12	0	12	0	12	**0**
G	6	12	18	12	18	**0**
H	11	18	29	18	29	**0**
J	2	29	31	34	36	5
K	5	0	5	17	22	17
L	7	5	12	22	29	17
M	5	12	17	29	34	17

Float calculation

Float is calculated as:

$$Float = LS - ES$$

or equivalently:

$$Float = LF - EF$$

For example, activity C:

$$Float_C = 8 - 0 = 8$$

Activity F:

$$Float_F = 0 - 0 = 0$$

4. Critical Path

An activity is **critical when its total float is zero**.

From the table:

- F → Float = 0
- G → Float = 0
- H → Float = 0
- E → Float = 0

Therefore:

$$F \rightarrow G \rightarrow H \rightarrow E$$

is the critical path.

Its duration is:

$$12 + 6 + 11 + 7 = 36$$

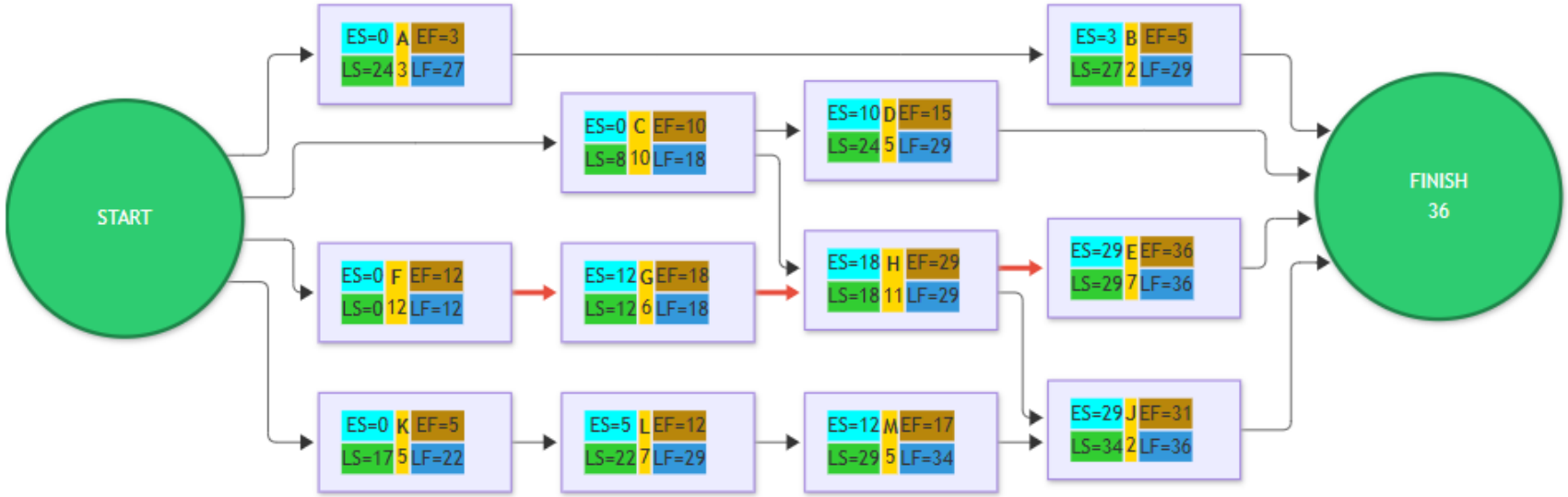
Therefore:

$$\text{Project Duration} = 36$$

5. Final ES/EF/LS/LF Network

The node format follows the lecture-style design:

```
┌──────────┬──────────┬──────────┐
│    ES    │ Activity │    EF    │
├──────────┼──────────┼──────────┤
│    LS    │ Duration │    LF    │
└──────────┴──────────┴──────────┘
```



Exercise 3 — Horizon Cable

Problem

Horizon Cable is expanding its cable TV service in Smalltown. The activities required to complete the expansion are:

Activity	Description	Immediate Predecessors	Duration
A	Choose stations	—	2 weeks
B	Get town council approval	A	4 weeks
C	Order converters	B	3 weeks
D	Install new dish to receive new stations	B	2 weeks
E	Install converters	C, D	10 weeks
F	Change billing system	B	4 weeks

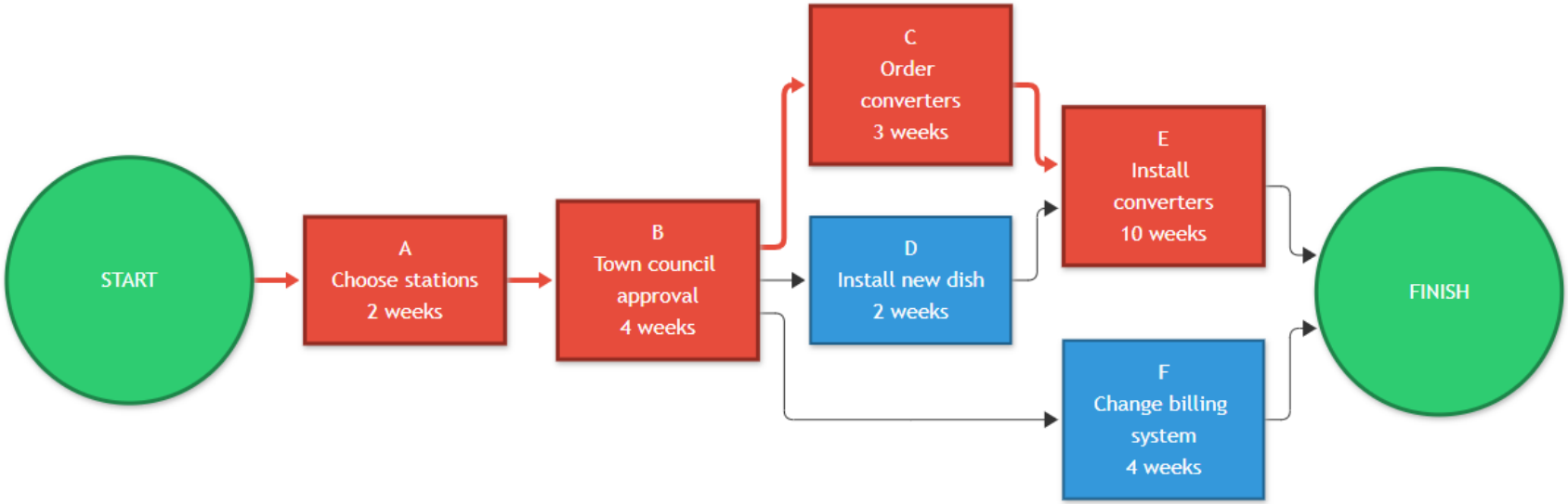
Precedence relationships

The network is:

- A → B
- B → C
- B → D
- B → F
- C → E
- D → E

Because **E requires both C and D**, E can only begin once both predecessor activities have finished.

Roy Diagram


START
A
Choose stations
2 weeks
B
Town council
approval
4 weeks
C
Order
converters
3 weeks
D
Install new dish
2 weeks
E
Install
converters
10 weeks
F
Change billing
system
4 weeks
FINISH

Path calculations

Path 1 — A → B → C → E

$$2 + 4 + 3 + 10 = 19$$

19 weeks

Path 2 — A → B → D → E

$$2 + 4 + 2 + 10 = 18$$

18 weeks

Path 3 — A → B → F

$$2 + 4 + 4 = 10$$

10 weeks

Float

The path-level slack is:

D branch

$$19 - 18 = 1 \text{ week}$$

Therefore, the D branch has **1 week of slack**.

F branch

$$19 - 10 = 9 \text{ weeks}$$

Therefore, the F branch has **9 weeks of slack**.

The critical-path activities have zero float:

$$A,\ B,\ C,\ E$$

Critical Path

The critical path is the longest path through the network.

Comparing the three paths:

Path	Duration
A → B → C → E	**19 weeks**
A → B → D → E	18 weeks
A → B → F	10 weeks

Therefore:

$$\text{Critical Path} = \text{A} \rightarrow \text{B} \rightarrow \text{C} \rightarrow \text{E}$$

and:

$$\text{Project Duration} = 19 \text{ weeks}$$

Project Definition — EcoMove Solutions

1. Introduction:

EcoMove Solutions is a company founded by Laura Gómez and two partners, specializing in the sale and installation of electric bicycle-sharing systems for municipalities and university campuses. The project aims to prepare the company for operational launch by coordinating business planning, location identification, contractual agreements, equipment procurement, office infrastructure installation, marketing preparation and the launch of promotional activities.

The project is scheduled to start on Monday, 5 October 2026. The working calendar is Monday to Friday, from 08:00 to 15:00, representing seven working hours per day, with weekends treated as non-working days. All three employees are on holiday in August.

The project team consists of Laura Gómez, Project Manager, at €35 per hour; Marc, Technical Specialist, at €22 per hour; and Clara, Administrative Assistant, at €12 per hour. The project plan comprises 12 work activities followed by a zero-duration operational launch milestone.

The planning exercise covers task sequencing, resource allocation, cost estimation, schedule development, critical path analysis, resource leveling and baseline monitoring. The objective is to establish a coordinated and financially transparent plan that supports the company's transition from initial preparation to operational launch.

2. Project Time Configuration and Task Register

2.1 Project Time Configuration

To ensure consistency with the project definition, the working calendar and employee holidays were configured at the project level in Monday.com.

- **Project working schedule — EcoMove Work Schedule:** Monday to Friday, from 08:00 to 15:00, providing seven working hours per day. Weekends are treated as non-working days.
- **Holiday schedule — EcoMove Off Schedule:** 1 August to 31 August, reflecting the annual holidays of all three employees during August.

The screenshots below show the working schedule and holiday configuration in Monday.com. These settings establish the time constraints used for project planning and scheduling.

Additionally, the group members' display names have been updated to match those used in the original project definition. This ensures consistency between the assignment documentation and the resource names displayed in monday.com, making resource identification and assignment verification easier.

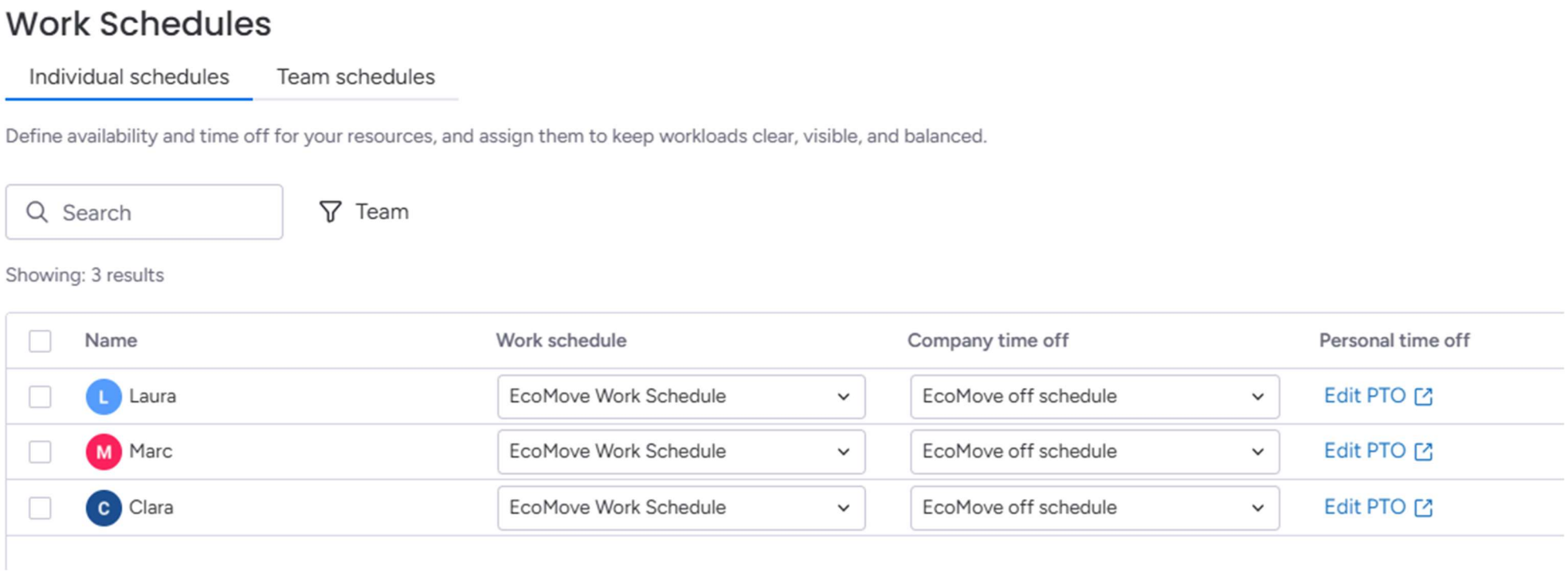


Figure 1: Project members display names and associated work and holiday schedules

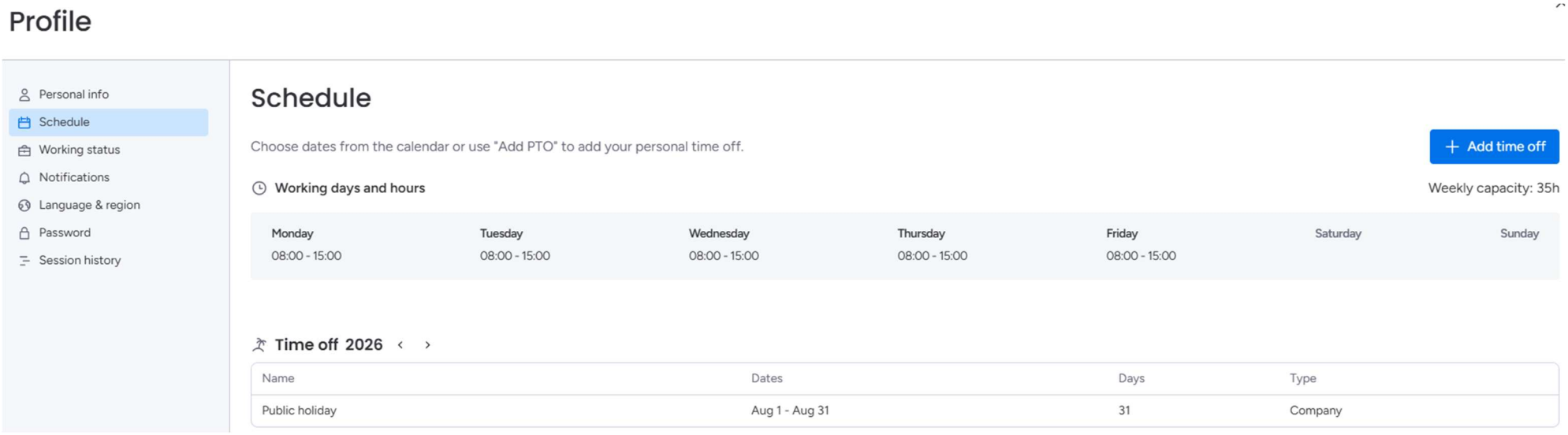


Figure 2: Details of Work and Absence schedules used in Monday.com

2.2 Task Register

The Task Register provides a structured overview of the 13 activities comprising the EcoMove Solutions project. It presents each task's identification code, description, configured duration and predecessor relationships, making the planned work sequence clear and traceable to the original assignment.

The register also reflects the resource-leveling adjustment to Task 2, distinguishing its original four-day duration from its revised five-day duration. To ensure readability and avoid omitting information, the graphical view was captured in three Landscape-oriented screenshots, allowing all columns to be displayed clearly.

All items 13 Items

Item	Group	Board	Owner	Status	Duration	Predecessor	Dependent On
13. Operational launch	Phase 3: Impleme...	EcoMove Solutions - Proje...		Not Started	0	12	12. Launch promotion...
1. Define business requirements	Phase 1: Planning ...	EcoMove Solutions - Proje...	L	Not Started	3	-	
5. Contact equipment suppliers	Phase 2: Procure...	EcoMove Solutions - Proje...	M C	Not Started	4	2	2. Prepare business pl...
8. Hire installation subcontractor	Phase 2: Procure...	EcoMove Solutions - Proje...	L C	Not Started	3	4	4. Sign lease agreem...
2. Prepare business plan	Phase 1: Planning ...	EcoMove Solutions - Proje...	L	Not Started	5	1	1. Define business req...
6. Purchase charging stations	Phase 2: Procure...	EcoMove Solutions - Proje...	M C	Not Started	2	5	5. Contact equipmen...
11. Sign municipal agreements	Phase 3: Impleme...	EcoMove Solutions - Proje...	L C	Not Started	3	9	9. Install office infrast...
9. Install office infrastructure	Phase 3: Impleme...	EcoMove Solutions - Proje...	M	Not Started	4	6,7	6. Purchase ch... +1
7. Purchase office equipment	Phase 2: Procure...	EcoMove Solutions - Proje...	M C	Not Started	2	4	4. Sign lease agreem...
4. Sign lease agreement	Phase 1: Planning ...	EcoMove Solutions - Proje...	L C	Not Started	1	3	3. Identify potential l...
3. Identify potential locations	Phase 1: Planning ...	EcoMove Solutions - Proje...	L	Not Started	5	2	2. Prepare business pl...
12. Launch promotional campaign	Phase 3: Impleme...	EcoMove Solutions - Proje...	L +2	Not Started	8	10,11	11. Sign munici... +1
10. Develop marketing materials	Phase 3: Impleme...	EcoMove Solutions - Proje...	L +2	Not Started	5	8	8. Hire installation su...

+ Add item

Figure 3: Task Register in Monday.com – part 1

All items

Item	Timeline	Baseline of Timeli...	Difference	Laura h	Marc h	Clara h	Laura Cost	Marc Cost
13. Operational launch	Nov 17	Nov 17	0	0	0	0	0	0
1. Define business requirements	Oct 5 - 7	Oct 5 - 7	0	18	0	0	630	0
5. Contact equipment suppliers	Oct 15 - 20	Oct 15 - 20	0	0	8	2	0	176
8. Hire installation subcontractor	Oct 23 - 27	Oct 23 - 27	0	4	0	4	140	0
2. Prepare business plan	Oct 8 - 14	Oct 8 - 14	0	32	0	0	1,120	0
6. Purchase charging stations	Oct 21 - 22	Oct 21 - 22	0	0	6	2	0	132
11. Sign municipal agreements	Nov 2 - 4	Nov 2 - 4	0	8	0	2	280	0
9. Install office infrastructure	Oct 27 - 30	Oct 27 - 30	0	0	16	0	0	352
7. Purchase office equipment	Oct 23 - 26	Oct 23 - 26	0	0	4	3	0	88
4. Sign lease agreement	Oct 22	Oct 22	0	2	0	1	70	0
3. Identify potential locations	Oct 15 - 21	Oct 15 - 21	0	10	0	0	350	0
12. Launch promotional campaign	Nov 5 - 16	Nov 5 - 16	0	4	20	4	140	440
10. Develop marketing materials	Oct 28 - No...	Oct 28 - No...	0	5	12	10	175	264

+ Add item

Figure 4: Task Register in Monday.com – part 2

All items

Item	Laura Cost	Marc Cost	Clara Cost	Total Labor Cost	Fixed Cost	Total Cost	EcoMoveApprov...	Baseline Differen...
13. Operational launch	0	0	0	0	0	0	Nov 17	0
1. Define business requirements	630	0	0	630	0	630	Oct 5 - 7	0
5. Contact equipment suppliers	0	176	24	200	0	200	Oct 15 - 20	0
8. Hire installation subcontractor	140	0	48	188	0	188	Oct 23 - 27	0
2. Prepare business plan	1,120	0	0	1,120	0	1,120	Oct 8 - 14	0
6. Purchase charging stations	0	132	24	156	8,000	8,156	Oct 21 - 22	0
11. Sign municipal agreements	280	0	24	304	0	304	Nov 2 - 4	0
9. Install office infrastructure	0	352	0	352	0	352	Oct 27 - 30	0
7. Purchase office equipment	0	88	36	124	0	124	Oct 23 - 26	0
4. Sign lease agreement	70	0	12	82	0	82	Oct 22	0
3. Identify potential locations	350	0	0	350	0	350	Oct 15 - 21	0
12. Launch promotional campaign	140	440	48	628	4,500	5,128	Nov 5 - 16	0
10. Develop marketing materials	175	264	120	559	0	559	Oct 28 - No...	0

+ Add item

Figure 5: Task Register in Monday.com – part 3

3. Dependency Structure and Gantt Chart

The Gantt Chart provides a visual representation of the EcoMove Solutions project schedule, showing the planned activities, their dates and the dependencies that determine their execution sequence. The dependency links reflect the predecessor relationships specified in the original assignment, including activities with multiple predecessors.

The critical path is highlighted in red and consists of Tasks 1 → 2 → 3 → 4 → 7 → 9 → 11 → 12 → 13. This sequence runs from defining the business requirements through business planning, location identification, lease signing, office equipment procurement, infrastructure installation, municipal agreements and the promotional campaign, culminating in the operational launch milestone. These activities form the critical sequence governing the planned project completion date.

A landscape-oriented screenshot of the Gantt Chart was included in the final report to maximize the readability of task names, schedule dates, dependency connectors and critical path highlighting. The displayed critical path matches the sequence identified in the project planning analysis.

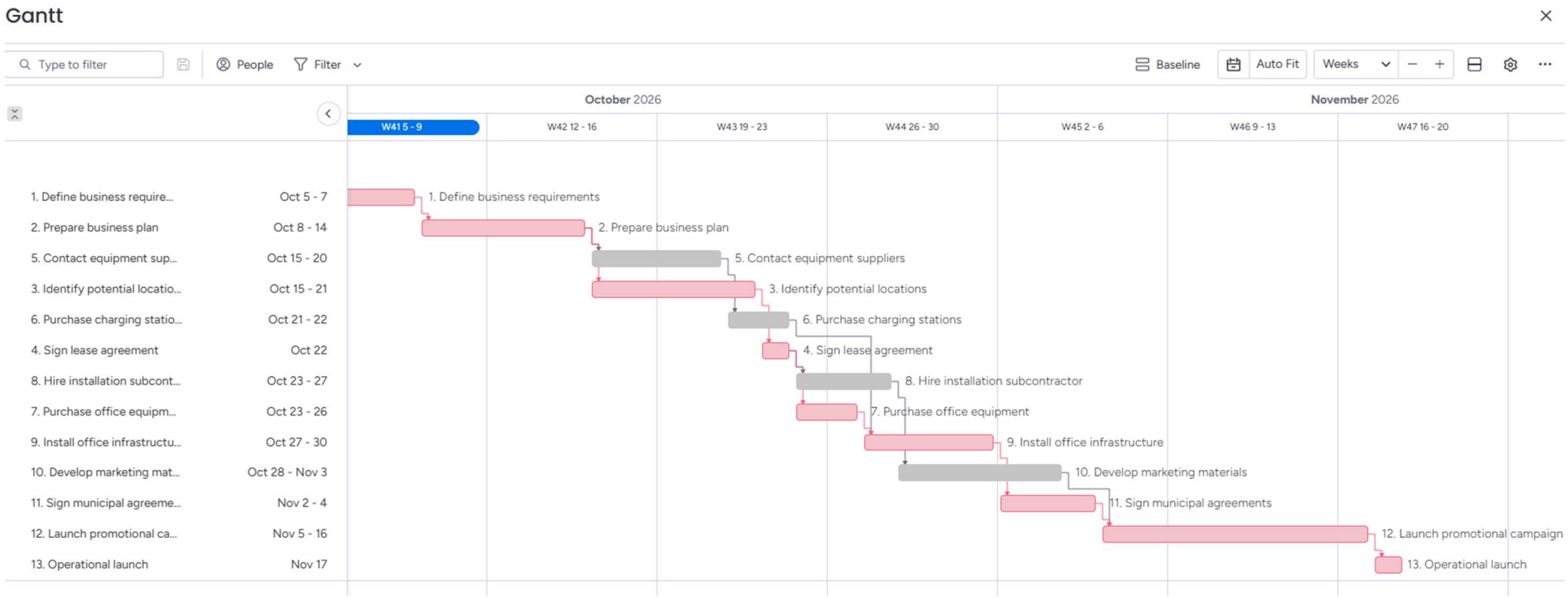


Figure 6: Gantt Chart view showing the tasks precedence and the critical path in red

4. Project Analysis and Key Findings

4.1 Project planning and schedule analysis

The EcoMove Solutions project plan integrates 12 work activities and a final operational launch milestone into a dependency-based schedule. The Gantt Chart identifies the critical path as Tasks 1 → 2 → 3 → 4 → 7 → 9 → 11 → 12 → 13, highlighting the sequence of activities that governs the planned completion date.

Resource leveling required extending Task 2, Prepare business plan, from four to five working days while retaining Laura's 32 assigned hours. This adjustment reduces her average assigned workload on that task from eight to 6.4 hours per working day, bringing it within the specified seven-hour daily capacity. The revised schedule preserves the assigned effort and associated labour cost.

4.2 Budget analysis

The total planned project budget is **€17,193**, comprising €4,693 in labour costs and €12,500 in fixed costs. Labour costs are calculated from the assigned hours and hourly rates of the three team members. The fixed costs comprise €8,000 for charging stations under Task 6 and €4,500 for the promotional campaign under Task 12. This breakdown provides a transparent financial estimate for the planned project activities.

4.3 Key project management challenges

The planning exercise addresses four main challenges:

1. **Schedule coordination:** Managing task dependencies and identifying activities that directly affect the planned completion date.
2. **Resource capacity:** Adjusting task duration to accommodate the team's working-hour constraints while preserving assigned effort.
3. **Cost control:** Combining labour and fixed costs to establish a consistent project budget.
4. **Future monitoring:** Using the approved baseline as a reference for identifying schedule changes and assessing deviations during execution.

5. Resource Planning and Assignments

Resource allocation matrix — assigned effort in hours

Code	Activity	Laura	Marc	Clara	Total
1	Define business requirements	18	0	0	18
2	Prepare business plan	32	0	0	32
3	Identify potential locations	10	0	0	10
4	Sign lease agreement	2	0	1	3
5	Contact equipment suppliers	0	8	2	10
6	Purchase charging stations	0	6	2	8
7	Purchase office equipment	0	4	3	7
8	Hire installation subcontractor	4	0	4	8
9	Install office infrastructure	0	16	0	16
10	Develop marketing materials	5	12	10	27
11	Sign municipal agreements	8	0	2	10
12	Launch promotional campaign	4	20	4	28
13	Operational launch milestone	0	0	0	0
Total assigned hours		**83**	**66**	**28**	**177**

The totals reconcile with the original assignment: Laura has 83 assigned hours, Marc has 66 hours and Clara has 28 hours, giving a total of 177 assigned hours.

Resource profiles and hourly rates

Resource	Role	Hourly rate	Assigned hours	Labour cost
Laura Gómez	Project Manager	€35	83	€2,905
Marc	Technical Specialist	€22	66	€1,452
Clara	Administrative Assistant	€12	28	€336
Total			**177**	**€4,693**

Labour costs are calculated by multiplying each resource's assigned hours by the corresponding hourly rate. For example, Laura's labour cost is 83 × €35 = €2,905. The total planned labour cost is €4,693.

6. Resource Allocation Analysis and Leveling

Resource allocation analysis evaluates the distribution of assigned working hours across the project schedule to identify potential resource overallocations. Resource leveling was applied to Task 2, *Prepare business plan*, extending its duration from four to five working days while retaining Laura's 32 assigned hours. This reduces her average workload on the task from 8 to 6.4 hours per working day, within the specified seven-hour daily capacity, without increasing the planned labour cost. The workload analysis and revised schedule provide a basis for assessing resource capacity and the impact of leveling on the project timeline.

7. Project Duration, Budget and Baseline Monitoring

7.1 Project duration

The EcoMove Solutions project is scheduled to start on 5 October 2026 and reach its operational launch milestone on 17 November 2026, according to the Gantt Chart. The schedule follows the specified Monday-to-Friday working calendar, with seven working hours per day and weekends treated as non-working days.

The critical path consists of Tasks 1 → 2 → 3 → 4 → 7 → 9 → 11 → 12 → 13, identifying the sequence of activities that governs the planned operational launch. Resource leveling extended Task 2, *Prepare business plan*, from four to five working days while preserving Laura's 32 assigned hours and keeping her average assigned workload on that task within the seven-hour daily capacity.

7.2 Project budget

The total planned project budget is €17,193, comprising €4,693 in labour costs and €12,500 in fixed costs. Labour costs are calculated from the assigned hours and hourly rates of the three project resources. Fixed costs comprise €8,000 for charging stations under Task 6 and €4,500 for the promotional campaign under Task 12.

This consolidated estimate provides a financial reference for the planned activities and supports cost monitoring during project execution.

7.3 Baseline and future monitoring

The project baseline provides a reference for monitoring the schedule as the project progresses. Comparing current task dates with the approved plan can help identify schedule changes and assess their potential impact on the planned operational launch.

The baseline information configured in Monday.com supports this monitoring process. The baseline representation and any available schedule-variance indicators should be interpreted according to the features actually verified in the project board.

8. Conclusion

The EcoMove Solutions project plan brings together the activities, dependencies, schedule, resource assignments and cost estimates required to prepare the business for operational launch. The Task Register and Gantt Chart provide a structured overview of the planned work, its sequencing and the critical path leading to the operational launch milestone on 17 November 2026.

The resource planning process consolidates 177 assigned working hours across Laura Gómez, Marc and Clara, with a total planned labour cost of €4,693. Resource leveling extended Task 2, *Prepare business plan*, from four to five working days while preserving Laura's assigned hours and reducing her average daily workload on that task to 6.4 hours.

The total planned project budget is €17,193, including €4,693 in labour costs and €12,500 in fixed costs. The project schedule and baseline arrangements provide a foundation for monitoring progress and identifying potential deviations during execution.

Overall, the planning exercise establishes a coordinated approach to project scheduling, resource allocation and cost control, supporting EcoMove Solutions in its preparation for operational launch.